Learning Objective 4.1: I can describe FSK modulation and implement an FSK transmitter and receiver in GNU Radio.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **Transmitter settings.** A flowgraph samples at $f_s = 48$ kHz and sends FSK at 1200 baud with a deviation of $\Delta f = 600$ Hz, one bit per symbol.

   (a) Find the samples per symbol to enter in the Repeat block.

   $$\text{sps} = \frac{f_s}{R_s} = \frac{48000}{1200} = 40$$

   40

   (b) Find the sensitivity to enter in the Frequency Mod block.

   $$k = \frac{2\pi\Delta f}{f_s} = \frac{2\pi \times 600}{48000} = 0.0785 \text{ rad/sample}$$

   0.0785

   (c) Find the modulation index and estimate the bandwidth with Carson's rule.

   $$h = \frac{2\Delta f}{R_s} = \frac{1200}{1200} = 1$$
   $$B \approx 2(\Delta f + R_s) = 2(600 + 1200) = 3.6 \text{ kHz}$$

   h = 1

   B = 3.6 kHz

   (d) A classmate builds the same transmitter from two Signal Sources at $+600$ and -600 Hz and a selector that switches between them at each bit. Explain why Frequency Mod gives a more compact spectrum.

   The two sources run independently, so at each switch the phase jumps from wherever one source is to wherever the other is. Those jumps are abrupt changes in the signal, and abrupt changes spread power across a wide band. Frequency Mod adds a phase step every sample, so the phase is continuous across each bit boundary and the spectrum stays compact.

2. **Receiver settings.** The receiver for the link of Question 1 follows the L8 flowgraph: Low Pass Filter, Quadrature Demod, Moving Average, Symbol Sync, and Binary Slicer.

   (a) Find the Quadrature Demod gain that puts the two tones at $+1$ and -1.

   $$g = \frac{f_s}{2\pi\Delta f} = \frac{48000}{2\pi \times 600} = 12.7$$

   12.7

   (b) Find the low-pass filter cutoff that passes the Carson bandwidth.

   The filter passes $-f_c$ to $+f_c$, so the cutoff is half of B:

   $$f_c = \Delta f + R_s = 600 + 1200 = 1800 \text{ Hz}$$

   1800 Hz

   (c) A classmate leaves the Quadrature Demod gain at 1. Find the levels the two tones produce, and explain whether the slicer still decides correctly.

   With $g = 1$ the output is the phase step itself, $\pm 2\pi\Delta f / f_s = \pm 0.0785$. The slicer compares with zero, and both levels are still on the correct side of zero, so the bits come out correctly. Only the scale is wrong: the eye diagram and any plot set for ± 1 show a nearly flat line.

   (d) The classmate also sets the filter cutoff to 8 kHz. By how many dB does that lower the SNR at the demodulator?

   The filter now passes 16 kHz of noise instead of 3.6 kHz, and the signal is unchanged:

   $$10 \log_{10}\left(\frac{16}{3.6}\right) = 6.5 \text{ dB}$$

   6.5 dB

3. **Noise.** The L8 link samples at 32 kHz, sends 1000 baud with $\Delta f = 1$ kHz, and filters to the 4 kHz Carson bandwidth. The signal has power 1, and the Channel Model's noise of voltage σ has power σ^2 spread over the full 32 kHz.

   (a) Find the SNR at the demodulator for $\sigma = 0.8$.

$$\text{SNR} = \frac{1}{\sigma^2 \times 4/32} = \frac{8}{0.64} = 12.5 = 11.0 \text{ dB}$$

   11.0 dB

   (b) The text starts to show errors below about 9 dB. Find the largest noise voltage that keeps the SNR at 9 dB.

$$\sigma^2 = \frac{8}{10^{0.9}} = 1.01, \qquad \sigma = 1.0$$

   $\sigma = 1.0$

   (c) The link doubles its symbol rate to 2000 baud and keeps $\Delta f = 1$ kHz. Find the new Carson bandwidth and the largest noise voltage for 9 dB.

$$B = 2(1000 + 2000) = 6 \text{ kHz}$$

$$\sigma^2 = \frac{32/6}{10^{0.9}} = 0.67, \qquad \sigma = 0.82$$

   B = 6 kHz

   σ = 0.82

   (d) Lesson 2 designs a link with margin above the SNR it needs. Explain, from what the L8 link does between 9 dB and 3 dB, why margin matters for FSK.

   The link goes from almost no errors at 9 dB to almost no packets at 3 dB, a range of about 6 dB. A link designed exactly at 9 dB therefore fails completely after a fade of a few dB. Margin keeps the SNR above the threshold when the path loss changes.

4. **An FSK remote on a spectrogram.** An RTL-SDR capture, viewed as a spectrogram in URH, shows a burst with two tracks, at 433.900 and 433.940 MHz. In the alternating preamble, the signal stays on each track for 0.417 ms.

   (a) Find the center frequency, the deviation, and the symbol rate.

$$f_c = \frac{433.900 + 433.940}{2} = 433.920 \text{ MHz}$$
$$\Delta f = \frac{40 \text{ kHz}}{2} = 20 \text{ kHz}$$
$$R_s = \frac{1}{0.417 \text{ ms}} = 2400 \text{ baud}$$

f_c = 433.920 MHz

Δf = 20 kHz

R_s = 2400 baud

   (b) The remote's crystal is accurate to 10 ppm. Find the largest frequency offset at 433.92 MHz.

$$10 \times 10^{-6} \times 433.92 \times 10^{6} = 4.3 \text{ kHz}$$

4.3 kHz

   (c) An FSK receiver tolerates an offset of about half its deviation. Explain whether a 4.3 kHz offset breaks the remote's link, and whether it would break the L8 link if that link ran at 433.92 MHz.

   The remote tolerates about 10 kHz, so 4.3 kHz is well inside its limit; the large deviation is what makes a crystal with a 10 ppm tolerance acceptable. The L8 link tolerates about 600 Hz, so a 4.3 kHz offset pushes both of its tones far off the receiver's filter and the link fails.

   (d) With a 600 Hz offset the L8 demodulator outputs levels near $+1.6$ and -0.4. Name one change to the receiver that would put them back at ± 1.

   Subtract the average of the demodulator output, for example with a DC Blocker after the Quadrature Demod, because the offset appears as a constant $f_{\text{off}}/\Delta f$ added to both levels. Measuring the offset and shifting the signal back in frequency before the filter also works.

Documentation: _____